

Assignment-5

1. Randomized Algorithm design using wrong-answer probability bound.

Yes, this knowledge can be used to design a useful randomized algorithm for F .

Algorithm Description: The key observation is that there exists a gap between probability of correct answer and any specific wrong answer.

$$\Pr[A(x) = F(x)] \geq 1/3 > 1/4 \geq \Pr[A(x) = \alpha] \quad \text{for any wrong } \alpha$$

We exploit this gap with the following algorithm:-

Algorithm $B(x)$:

1. Run $A(x)$ independently K times, producing outputs $r_1, r_2, \dots, r_K$
2. Output any value that appears more than $K/4$ times among the results.
3. If no such value exists, output an arbitrary default value.

Correctness analysis:

Let X_c denote the number of times the correct answer $F(x)$ appears across K runs, and let X_α denote number of times any specific wrong answer α appears. By linearity of expectation:

$$E[X_c] \geq K/3, \quad E[X_\alpha] \leq K/4$$

The gap between these expectations is $K/3 - K/4 = K/12$

We want to show that with high probability, $F(x)$